Module 03

Khansa Rubab

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Materials and Method

Title: Alignment of DNA sequence to reference genome

Whole-genome sequencing (WGS) reads obtained from breast cancer samples were aligned to the human reference genome, with a particular emphasis on chromosome 19, using Geneious Mapper v9.1.8. The alignment was performed with a low-sensitivity setting to strike a balance between computational efficiency and accuracy, resulting in the generation of contigs for subsequent analyses.

Results

Title: Analysis of Reads Mapping to Reference Genome or Reads Alignment or Reads Mapping:

In our study, utilizing Geneious software, 95,130 out of 2,312,026 reads were assembled to chromosome 19 to produce producing single contigs with a total length of 58,631,252 bases, exhibiting 93.4%% identical sites and also pairwise % identity of 93.4%; the mean coverage was 0.3x with a standard deviation of 2.1 , while the minimum and maximum coverage were0x and 1400, respectively, and the reference genome coverage was 12.8% (7,510,251 of 58,617,616bp); the mean Ungapped length of reads was 48.0 bases with a standard deviation of 2.0

, and the base composition consisted of 25.5%Adenine, 24.2% Cytosine, 24.3%

Guanine, and 25.7%Thymine, resulting in a GC content of 48.6%; no indels were detected, and the molecular weights were calculated as 2.39×10^7 kDa for single-stranded DNA and 4.784×10^7 kDa for double-stranded DNA

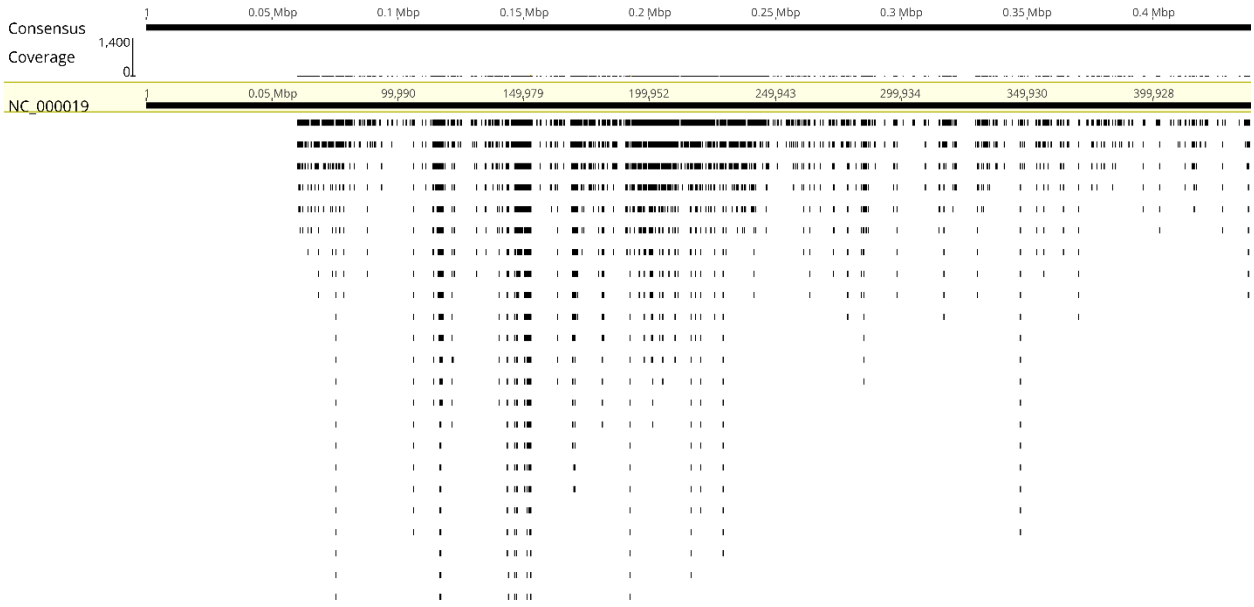


Figure 1: Aligned reads from a Breast cancer dataset mapped to human Chromosome 19.